

Sweeney → EXAM 4

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① given: $T = 625$, $t = 500$ d, $P_{O_2} = 1.8$, $P_{Zr} = 6.5$, $P_{ZrO_2} = 5.68$ 1

a.) $t^* = 6.62 \cdot 10^{-7} \exp\left(\frac{11949}{T}\right) = 6.62 \cdot 10^{-7} \exp\left(\frac{11949}{625}\right)$

$t^* = 133$ DAYS. since $t > t^*$, use $\delta = k_L t^{1/n}$

Find $\delta^* \rightarrow \delta^* = 5.1 \exp\left(\frac{-550}{625}\right) = 2.11 \mu m$

$K_L = 7.48 \cdot 10^6 \exp\left(\frac{-12500}{625}\right) = 0.0154$

Finally, $\delta = k_L t^{1/n}$ -only before t^* Assume $n=4$

$\delta = \delta^* + k_L(t - t^*)$

$\delta = 0.0154 \cdot (500)^{1/4} = 0.0728$

$\therefore \delta = \delta^* + \delta = 2.11 \mu m + 0.0728 \mu m$

$\delta = 2.18 \mu m$

b.) $f = 0.14$, $t^0 = 600 \mu m$, $P_{ZrO_2} = 5.68$, $P_{Zr} = 6.5$

$$C_H = \frac{2 \cdot f \cdot \delta \cdot P_{O_2} \cdot f_{ZrO_2} + \frac{m_H}{m_O}}{\left(t - \frac{\delta}{P_{O_2}}\right) \cdot P_{metal}} \quad \left| \quad f_{ZrO_2} = \frac{16 \cdot 2}{16 \cdot 2 + 91} = 0.2601 \right.$$

$\frac{m_H}{m_O} = 1/16$ $P_{O_2} = 1.56$

$C_H = \frac{2 \cdot 0.14 \cdot 2.18 \mu m \cdot 0.2601 \cdot \frac{1}{16} \cdot 5.68}{\left(600 - \frac{2.18}{1.56}\right) \cdot 6.5}$

$C_H = 14.5 \text{ ppm}$

② Oxygen ion diffusion to the metal-ox interfacing point 2

③ Form at outer edge of cladding, where it is cooler. This is because H is going to ~~move~~ travel according to the temperature gradient. *- more here*

Impacts of hydrides $\rightarrow$ worsened fracture resistance, ductility
 $\rightarrow$ hydride cracking may occur
 $\rightarrow$ reduced mechanical integrity overall

④ Reactivity Initiated Accident, a rapid ~~reactivity~~ insertion of reactivity. LWR $\rightarrow$ PWR $\rightarrow$ control Rod ejection
 $\rightarrow$ BWR $\rightarrow$ control Rod drop

The power increase from the reactivity insertion causes a lot of energy to go to fuel. Therefore, rise in fuel temperatures and associated effects like thermal exp. Fuel pellets may swell and cause cladding to fail. *- needed more*

⑤ Loss of Coolant Accident - usually a break in RCPB causes inventory loss & thereby diminished cooling capability.

Loss of heat removal causes cladding to heat up, which then leads to ~~oxidation~~ oxidation as cladding interacts with steam. Cladding may balloon due to new thermal condition and could rupture. *- wanted a bit more*

Main Differences

$\rightarrow$ RIA caused by power excursion, LOCA caused by loss of cooling.

In the fuel, brittle fracture is usually seen in RIAs while ballooning is more representative of LOCA (though not exclusive)

⑥ higher burnup fuel will have more hydride Accumulation and therefore more susceptible cladding due to embrittlement.

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high Burnup fuel would also have a thicker oxide layer and worse thermal conductivity.

Generally, higher burnup fuel has a larger failure probability.

Both effects above would mean high B fuel is more susceptible to RIA (PCMI) ; LOCA (brittle fracture) failure mechanisms.

⑦ 1. Use of Ceramic fuel / composite fuel

2. Coatings for Zr cladding may slow oxidation

3. Cladding Alloys vs. traditional cladding (e.g. Silicon Carbide)

4. Changing fuel type to have lower k_p , therefore less stored heat
- related, but not what I was looking for

⑧ Zr Coatings, like FeCrAl. It is minimally invasive to the existing fuel ~~reactor~~ system Architecture, may reduce oxidation during LOCA or steam-broke transient. Improves mechanical props without great performance detriment.

FeCrAl not a coating, a replacement $\rightarrow H_2O + Zr \rightarrow ZrO_2 + H_2$

⑨ generates heat thru Oxidation Reaction, leading to temp. rise. Hydrogen created during oxidation is also a concern with potential hydrogen detonation.

Ultimately, cladding becomes more brittle due to oxygen intrusion, therefore more susceptible to failure

- wanted more

⑩ 1. Boiling (film boiling/DNB) -

2. Fuel Temp Limits during operation & cooling need / Fuel melting

3. Coolant Chemistry (pH, Zn, O₂) - not really -

⑪ CRUD causes a coating to deposit onto the cladding surface, which can ~~also~~ diminish the fuel's heating. Also, CRUD can fall off of cladding and gather on other surfaces or within other systems (i.e. could fail a filter, for example). CRUD may also become activated due to irradiation & cause ~~more~~ ^{unstable} radionuclides to form.

⑫ IN PWR'S → boric acid is used as a chemical shim for reactivity. - this is not a chemistry measure, but something to be counter-acted

IN BWR'S, Zinc injection OR On-Line Noblechem. Zn injection causes the formation of a protective film on stainless steel components, can protect IGSCC.

ONLC mitigates IGSCC by reacting H₂O₂ using noble metal catalysts. Also prevents hydrogen ~~detrit~~ ^{detrimental} formation.

⑬ 1. hot spots caused by some heterogeneity in the fuel. Pu-richer regions will have greater power density and therefore higher temps.

2. Fission gas release risk → fission gas may release quicker in the high Pu-zones described above. The fission risk is higher in these sections, therefore higher internal pressures within the rod must be accounted for during fuel design.

- two parts of same concept